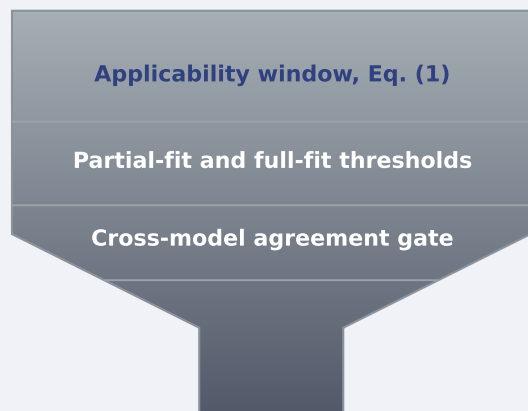


Heterogeneous $J_c(T,H)$ literature



$n = 934$ articles retrieved and screened through the Elsevier and Springer APIs; fitted curves also come from arXiv preprints and NHMFL records (Sec. II.A)



Cohort A: $J_c(T)$ at fixed H

Cohort B: $J_c(H)$ at fixed T

$T/T_c < 0.7$

$\Delta H/H_{c2} > 0.3$

- 50** papers contributing fitted curves
- 35** compound labels, 27 composition keys
- 3303** extracted critical-current points
- 20** compounds fittable on both axes
- 257 / 52** β_T / β_H fits, the latter from 12 papers
- 96** per-paper anchors over 32 papers

Multi-stage substructure conditional aggregation

the conditioning claim is a temperature-axis claim

on the field axis the family label accounts for 0.038 of the between-paper variance in the exponent, at $p = 0.98$, and no family-level verdict is reported for it (Table III)

Stage 1	Stage 2	Stage 3
monolithic regression on one descriptor; rank scope only	sample-form conditional median inside each substructure	substructure aggregate median with an explicit IQR bound

$$\log_{10} J_c = \log_{10} J_{c, \text{partial}} + \beta \log_{10}(1 - x/x_c)$$

fitted per axis: β_T separately, β_H separately

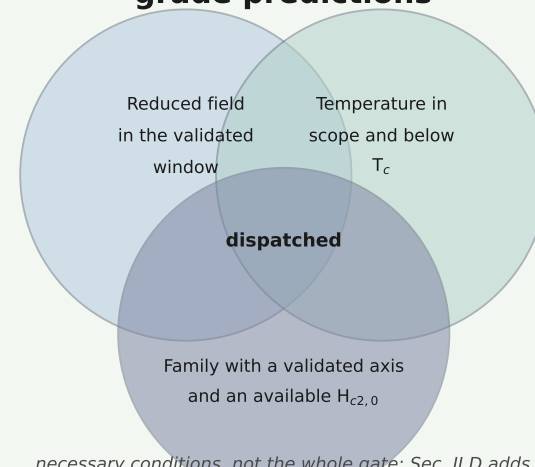
No fold improvement is reported.

Under leave-one-substructure-out the errors are 12.3 without sample-form conditioning and 14.9 with it over seven families, 1.19 and 0.55 on the fits passing physicality over three, and removing any one family moves the comparison across unity.

The sample-form diagnostic sets the rule

the predictor follows. After the repair of Sec. III.F it explains 0.81, 0.37 and 0.04 of the within-family scatter in the anchor, for the 11-type, 122-type and MgB_2 -class families in turn. It does not establish the distinction to significance: sample form is nearly a relabelling of source paper here, and under the only null that respects that structure the MgB_2 cell can never return p below 0.33. *Two regimes, reported as the rule and not as a finding.*

Bounded screening grade predictions



necessary conditions, not the whole gate: Sec. II.D adds anchor density, monotonicity and population screens

Universal (T/T_c , H/H_{c2}) scaling is not adopted

within-bin standard deviation falls 13.0%, 24.3% on the variance scale, both below the 30% adoption threshold, on the pre-withdrawal 23-compound cohort (Table III)

Family-scope $J_c(T,H)$ envelopes

with a 95% bootstrap interval and a refusal flag

- 183** candidate compounds evaluated
- 84** dispatched, every one MgB_2 -class
- 60** refused a field target, no $H_{c2,0}$ anchor
- 4.98** $\log_{10} J_c$ at 4.2 K and 5 T

One family curve per grid point, shared by every member candidate, and no per-compound ranking is emitted. The 60 keep their temperature-axis output (Sec. III.E).